



Experimental Investigation of Pile Group Effects and Mitigation of Surcharge-Induced Negative Skin Friction Using Isolation Piles

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Abstract: Negative skin friction induced by surcharge loading represents a critical geotechnical issue, compromising the performance of pile foundations and associated superstructures. Although surcharge-induced negative skin friction has been extensively investigated for single piles, the underlying pile group effect and the effectiveness of isolation piles in mitigating dragload development remain insufficiently understood. To address these gaps, controlled laboratory model tests were conducted on a single pile and a 3×2 pile group subjected to surcharge loading. Axial strains along the pile shafts and pile-head displacements were continuously monitored to quantify the distribution characteristics of negative skin friction, evaluate pile group interactions, and assess the influence of isolation pile spacing on the reduction of negative skin friction acting on engineering piles. The results demonstrated that a pronounced pile group effect was generated under surcharge loading. Relative to the single-pile condition, the average negative skin friction acting on corner piles and center piles was reduced by 10.04% and 15.36%, respectively, indicating that the center pile was more strongly influenced by the pile group effect. Furthermore, the installation of isolation piles was shown to effectively decrease pile-head displacement, pile-shaft stress, and surcharge-induced negative skin friction. The mitigation efficiency was found to be strongly dependent on isolation pile spacing, with closer spacing producing a more pronounced shielding effect. As the spacing between isolation piles and engineering piles was reduced from $6D$ to $2D$, the reduction rate of average negative skin friction increased from 7.55% to 20.13%. However, the incremental improvement became progressively less significant when the spacing was smaller than $4D$. These findings provide experimental evidence for the pile group behavior of surcharge-induced negative skin friction and offer practical guidance for the design and optimization of isolation-pile systems intended to mitigate dragload effects in pile-supported structures.

Keywords: Surcharge loading; Negative skin friction; Pile group effect; Isolation pile; Laboratory model test

1 Introduction

Pile group foundations possess excellent stability and are widely applied in engineering projects such as bridges, wharves, and embankments [1, 2]. Single-sided surcharge loading induces soil settlement and horizontal displacement within a certain area, subjecting the pile foundation to the coupled action of negative skin friction and horizontal loading [3]. The negative skin friction around the piles can cause differential settlement of the foundation, while the horizontal load increases the bending moment of the pile shaft, which may ultimately lead to damage to both the pile foundation and the superstructure [4, 5]. Therefore, it is of great significance to investigate the pile group effect of negative skin friction under surcharge loading, as well as the influence of isolation pile layout patterns on the mitigation of negative skin friction.

Many scholars have conducted numerical simulations and physical model studies to investigate the negative skin friction of pile foundations. Regarding the negative skin friction mechanism in single piles, Wu et al. [6] investigated the influence of surface surcharge and consolidation time on the neutral plane location and dragload of a single pile based on the effective stress method and finite element numerical simulations. Hou et al. [7] conducted centrifuge model tests to realistically simulate the evolution of negative skin friction on a single pile induced by the self-weight

consolidation of backfill, analyzing the migration law of the neutral plane caused by negative skin friction and the nonlinear effect of relative pile-soil displacement on the interface friction strength. With the development of long-term field monitoring technologies, an increasing number of scholars have obtained valuable field data based on real engineering scenarios. Chalajour and Blatz [8] combined field tests with a unified design approach to evaluate the long-term calculation accuracy for foundation piles subjected to negative skin friction induced by soil consolidation settlement.

Through large-scale field tests, Xing and Liu [9] systematically verified the dynamic evolution path of negative skin friction along the pile shaft during the severe large vertical displacement of soil. These field-scale studies provide essential support for the unified assessment of negative skin friction under complex conditions. Ma et al. [10] comparatively analyzed the stress differences and variations in the neutral plane within a 3×3 pile group under various forms of large-area uniform surcharge loading. Yan et al. [11] conducted scaled model tests on pile group foundations under uniform fill surcharge, revealing the shielding effect that reduces the dragload on individual piles at different locations within the group. Alvi and Rahardjo [12] utilized a three-dimensional finite element method to comprehensively quantify the negative skin friction reduction factor for end-bearing pile groups under uniform surcharge loading. In recent years, increasing attention has been focused on the negative skin friction of pile groups subjected to adjacent surcharge loading. Through sand foundation model tests, Huang et al. [13] found that for pile groups under adjacent surcharge loading, when the pile spacing reaches five times the pile diameter, the pile group effect is no longer significant. Through numerical simulations, Guo et al. [14] investigated the effects of asymmetric surcharge loading on the deformation and mechanical responses of pile groups, pointing out that asymmetric loading leads to a highly asymmetrical distribution of lateral earth pressure and structural horizontal displacement.

To further quantify the nonlinear coupling relationship between the lateral earth pressure and the horizontal and vertical displacements of the pile, Zhou et al. [15] deeply explored the mechanical and deformation characteristics of passive piles under surcharge loading based on the strain wedge model. It was pointed out that under lateral soil squeezing, the pile shaft exhibits a complex response of "combined translation and rotation," and the additional earth pressure on the pile shaft is highly modulated by the relative pile-soil displacement. The aforementioned studies primarily focus on single piles and pile foundations under uniform surcharge loading; research on the negative skin friction of pile groups subjected to adjacent surcharge (asymmetric loading) remains relatively scarce. Mitigating the adverse effects of negative skin friction on pile foundations is another research hotspot. Common mitigation methods in engineering practice include bitumen coatings [16] and isolation casings [17]. However, these methods have limitations in resisting the horizontal lateral earth pressure induced by adjacent surcharge loading. Under such circumstances, a more effective approach is to install isolation piles between the surcharge and the engineering piles. Regarding experimental research, Huang et al. [18] investigated the practical effectiveness of the isolation pile method in reducing surcharge-induced negative skin friction on pile foundations through laboratory single-pile model tests, confirming its efficacy in settlement control. Liu et al. [19] conducted 1-g physical model tests to evaluate the barrier efficacy of differently arranged isolation piles against soil deformation and their impact on adjacent structures under asymmetric surcharge loading.

By combining centrifuge testing and three-dimensional numerical analysis, Liang [20] successfully quantified the shielding and barrier mechanisms of isolation piles against the dragload on protected pile groups during soft soil consolidation. This verified that peripheral isolation pile rows can effectively intercept soil deformation, providing cutting-edge theoretical support for the load-mitigation efficacy of isolation piles. In terms of numerical simulations and theoretical analyses, Yuan et al. [21] defined a dragload shielding effect coefficient based on a three-dimensional finite element model, providing a preliminary investigation into how isolation pile layouts (e.g., pile length and spacing) shield pile groups from negative skin friction under surcharge loading. Cao et al. [22] established a simplified continuum elastic analytical model to analyze the restraining effect of isolation piles on vertical soil displacements caused by unbalanced loading, revealing the relative slip characteristics at the pile-soil interface. Tao et al. [23] combined field measurements with three-dimensional numerical analyses to dissect the microscopic barrier mechanism of isolation piles in intercepting soil rupture surfaces and stress transmission paths. Although isolation piles have been utilized to mitigate negative skin friction, their mitigation mechanisms and the influence of layout configurations on the mitigation effectiveness remain largely unclear.

In summary, current research on the pile group effect of surcharge-induced negative skin friction and the mitigation efficacy of isolation piles remains relatively scarce. This study designs and conducts laboratory model tests in soft clay to investigate the negative skin friction of pile groups and the performance of isolation piles under surcharge loading. The objective is to examine the pile group effect of surcharge-induced negative skin friction and the influence of isolation pile spacing on the mitigation of negative skin friction on engineering piles.

2 Model Test

2.1 Experimental Design

In this study, a total of five sets of pile foundation model tests under surcharge loading in clay were designed. The specific experimental conditions are detailed in Table 1. By comparing Test 1 and Test 2, the pile group effect of negative skin friction under surcharge loading was investigated. By comparing Tests 2 through 5, the influence of isolation pile spacing on the load-bearing response of the engineering piles was explored. A 1-g physical model test was employed in this study. Taking into account the boundary effects of the test box and the dimensions of typical full-scale pipe piles (e.g., 600 mm in diameter), the outer diameter of the model pile was set to 30 mm, yielding a geometric scaling ratio of 1:20. Since 1-g tests cannot proportionally scale up the gravitational field, the stress level in the model ground is inevitably lower than that of the engineering prototype, and some stress-dependent properties of the soil may differ from in-situ conditions. Therefore, the findings of this experiment are primarily used to qualitatively reveal the mechanism of negative skin friction on pile groups under adjacent surcharge loading and to evaluate the relative load-mitigation efficacy of isolation piles.

Table 1. Summary of experimental conditions for model tests

Test No.	Loading Condition	Isolation Pile Layout	Engineering Pile Type	Engineering Pile Layout
1	Single-sided surcharge (15 kPa)	Without isolation piles	Straight pile	Single pile
2		Without isolation piles		3 × 2 pile group
3		Plexiglass pipe pile, $D = 30$ mm, pile spacing $2D$		
4		Plexiglass pipe pile, $D = 30$ mm, pile spacing $4D$		
5		Plexiglass pipe pile, $D = 30$ mm, pile spacing $6D$		

As illustrated in the plan view of the experimental setup in Figure 1, the tests were conducted within a custom-built steel test box. The width of the surcharge loading was 300 mm, and the distances from the edge of the surcharge to the isolation piles and the engineering piles were D and $4D$, respectively. To avoid boundary effects, the clearance between the engineering piles and the boundaries of the test box was strictly kept greater than $10D$. Figure 2 presents the cross-sectional view of the test box. From top to bottom, the experimental soil profile consisted of saturated sand, saturated clay, and saturated sand. The physical and mechanical parameters of the soils are detailed in Table 2. To ensure a fully saturated state, the clay was thoroughly mixed and soaked in water for one week prior to the tests.

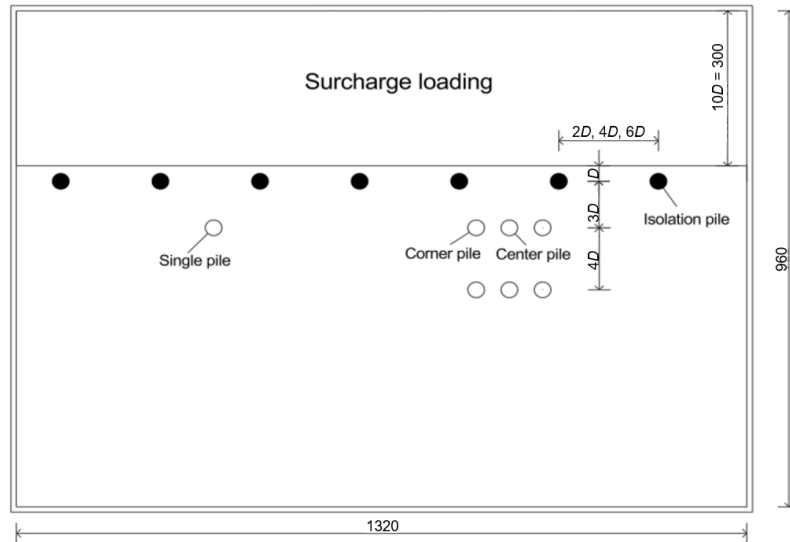


Figure 1. Plan view of the experimental setup (Unit: mm)

The model piles used in the tests (including both the engineering and isolation piles) have an outer diameter (D) of 30 mm, a length (L) of 1000 mm, and a wall thickness (t) of 2 mm. All piles are made of smooth plexiglass. Based on material tensile tests, the elastic modulus (E_{PG}) of the plexiglass was measured as 3.2 GPa, and its Poisson's ratio was 0.387.

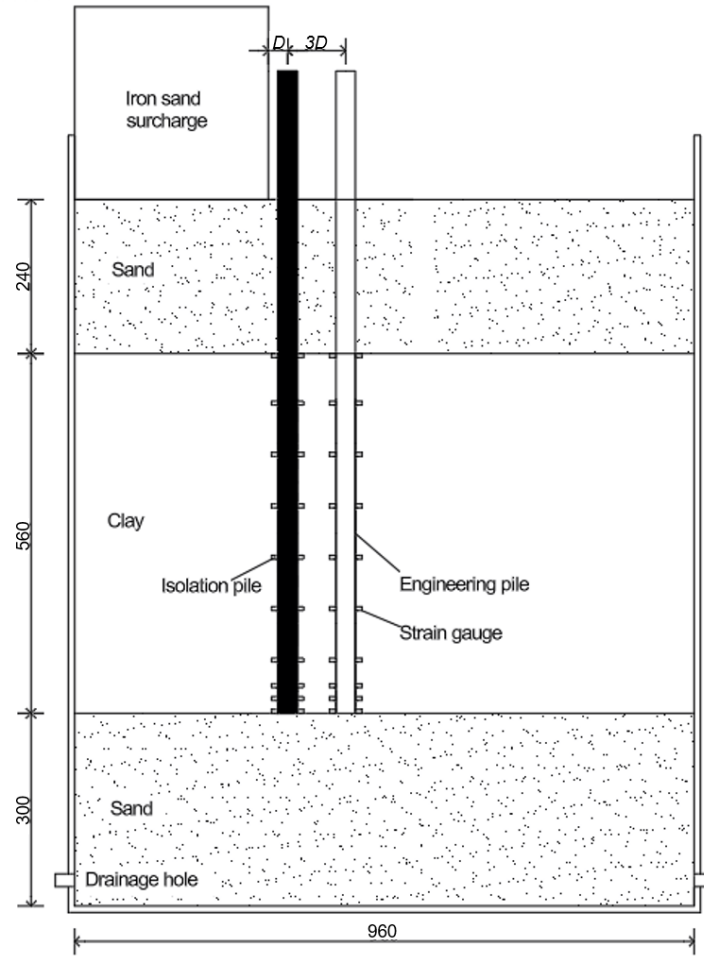


Figure 2. Cross-sectional view of the test box (Unit: mm)

Table 2. Soil parameters

Material	Cohesion (kPa)	Internal Friction Angle (°)	Permeability Coefficient (cm/s)	Plastic Limit (%)	Liquid Limit (%)	Max/Min Dry Density (g/cm ³)	Optimum Moisture Content (%)	Density (g/cm ³)
Clay	20.2	15.10	2.10×10^{-7}	19.60	39.20	1.682	16.20	2.74
Sand		21.50	1.60×10^{-3}			1.725/1.573		2.67

2.2 Test Installation

The basic experimental procedures included preparation of the model piles, placement of the bottom sand layer, fixation of the model piles, placement of the clay and top sand layers, and the installation and connection of testing equipment. To ensure consistent conditions across the five sets of tests, the operational steps were strictly conducted below. First, the sand was uniformly filled in layers by weight to achieve the pre-set relative density. Once the sand reached the designated height, a layer of geotextile was laid, and water was injected to saturate the sand. Next, the model piles were fixed in position, ensuring that the pile tips rested on the surface of the bottom sand layer. Subsequently, the saturated clay layer was slowly and evenly placed, followed by the placement of the top sand layer in stages, which was also saturated with water. The lead wires from the pile shaft strain gauges were then connected to the data acquisition system, and dial gauges were installed. Upon completion of these steps, the entire setup was left to stand for 48 hours to allow the soil to stabilize. Finally, the surcharge loading weights were applied, the drainage valve at the bottom of the test box was opened, and experimental data collection commenced.

2.3 Test Loading and Measurement

Iron sand was piled within a loading box to simulate the application of the surcharge loading. A flexible bag was placed inside the loading box, and the natural bulk density of the iron sand within the bag was 5.05×10^3

kg/m³. The final load applied to the soil surface was 15 kPa over an application area of 0.396 m². In engineering practice, a 15 kPa surcharge corresponds to an ordinary soil fill of approximately 0.8 to 1.0 m in height (assuming a unit weight of 18–20 kN/m³), which broadly represents common loading conditions such as land reclamation behind high-piled wharves or bridge abutment backfilling [24]. Displacements were measured using dial gauges, and pile shaft strains were measured using strain gauges, from which the pile shaft axial forces were subsequently calculated via conversion. Foil-type strain gauges with a resistance of 120 Ω , whose sensitivity met the experimental requirements, were utilized. Their layout locations are illustrated in Figure 3. After the test commenced, data was recorded every 10 minutes during the first 5 hours; thereafter, the sampling interval was adjusted to 1 hour.

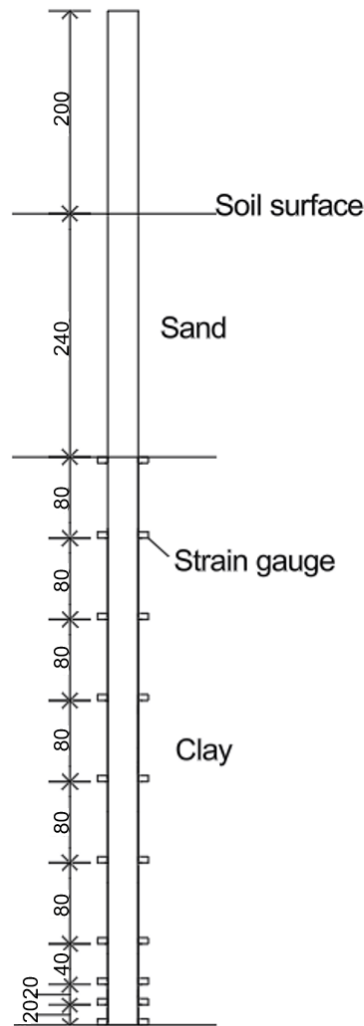


Figure 3. Layout of strain gauges on the pile shaft

3 Experimental Results and Analysis

3.1 Pile Group Effect under Surcharge Loading

The pile-soil displacement characteristics and the load-bearing characteristics of the pile shafts for a single pile (Test 1) and a 3×2 pile group (Test 2) without isolation piles were comparatively analyzed to investigate the pile group effect of negative skin friction under surcharge loading. Additionally, the pile group effect coefficient was employed to compare the intensity of the pile group effect on the corner piles versus the center piles within the group.

3.1.1 Influence of the pile group effect on pile-head displacement

Figures 4a and 4b, respectively, illustrate the variations of pile-head settlement and horizontal displacement with consolidation time for the single pile, as well as for the corner and center piles within the pile group. Compared to the single pile, the development trends of pile-head settlement and horizontal displacement for both the corner and center piles in the group are slightly more gradual. The pile group effect coefficients for the pile-head settlement of the corner and center piles are 0.94 and 0.90, respectively; the pile group effect coefficients for horizontal displacement

are 0.98 and 0.92, respectively. Because the corner pile is adjacent to fewer piles, it is less affected by the pile group effect than the center pile. Consequently, the soil displacement around the corner pile is larger, the earth pressure exerted on its shaft is greater, and the resulting dragload on the pile is higher. This causes the pile group effect coefficients for both the settlement and horizontal displacement of the corner pile to be greater than those of the center pile.

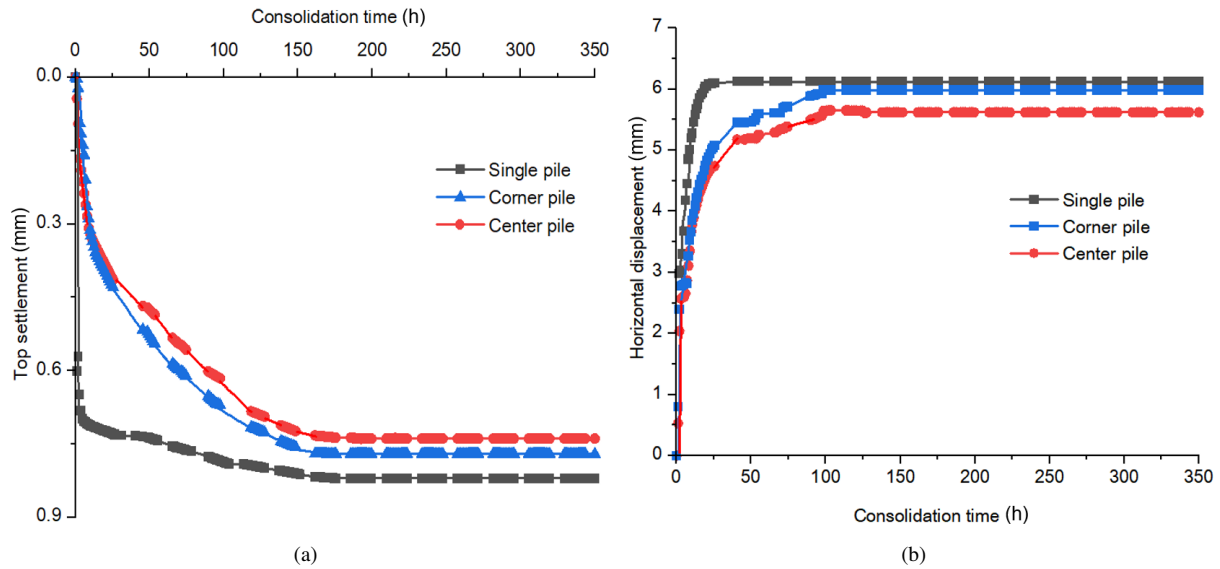


Figure 4. Influence of pile group effect on axial force and bending moment: (a) Pile top settlement; (b) Pile horizontal displacement

3.1.2 Influence on the pile shaft forces of the center pile in the group

(i) Axial force and bending moment

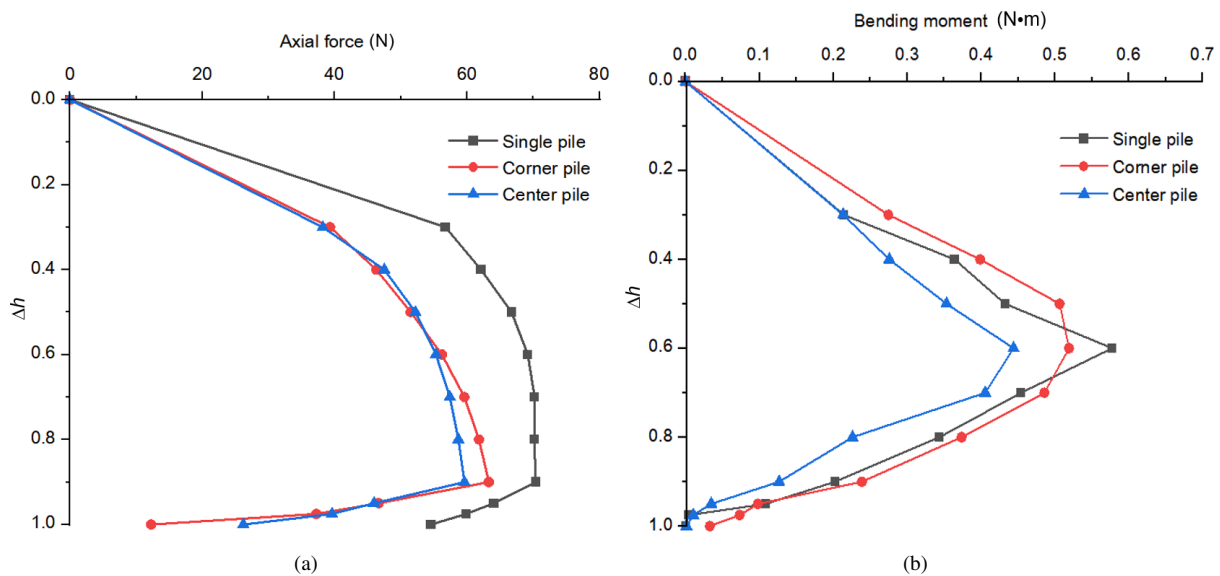


Figure 5. Influence of the pile group effect on axial force and bending moment: (a) Axial force profile; (b) Bending moment profile

Figures 5a and 5b illustrate the distributions of axial force and bending moment along the relative depth (Δh , the ratio of measurement depth to pile length) for the single pile, as well as for the corner and center piles within the pile group. The pile shaft axial force first increases and then decreases, reaching its maximum value at a relative depth of 0.9 for all cases. The pile group effect coefficients for the axial forces of the corner and center piles are 0.9 and 0.85, respectively. Similarly, the pile shaft bending moment first increases and then decreases with relative

depth, with the pile group effect coefficients for the bending moments of the corner and center piles being 0.9 and 0.77, respectively. The pile group effect has a more pronounced impact on the axial force and bending moment of the center pile. This is because the corner pile is only influenced by adjacent piles on one side, whereas the center pile is influenced by adjacent piles on both sides. The presence of surrounding piles reduces the additional stress acting on the soil around the pile itself. Consequently, the surcharge effect on the piles within the group is lower than that on the single pile, and the surcharge effect on the center pile is lower than that on the corner pile.

(ii) Neutral plane

During the experiment, the load applied to the pile shaft was relatively small, resulting in minimal compression of the pile shaft. Therefore, the settlements at the pile head and pile tip were considered to be identical. By combining the measured settlements of the pile head and the soil surface (s), the pile-soil settlement profile for the single pile was obtained, as shown in Figure 6. The neutral planes for the single straight pile, as well as for the corner and center piles within the group, are all located at a relative depth of approximately 0.9, which coincides with the location of the maximum axial force observed in the pile shaft during the experiment.

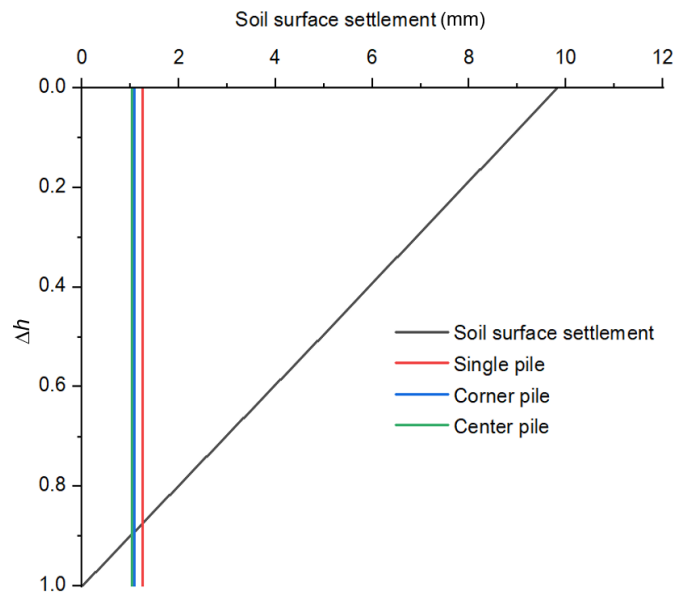


Figure 6. Pile-soil settlement

3.1.3 Influence of pile group effect on average negative skin friction

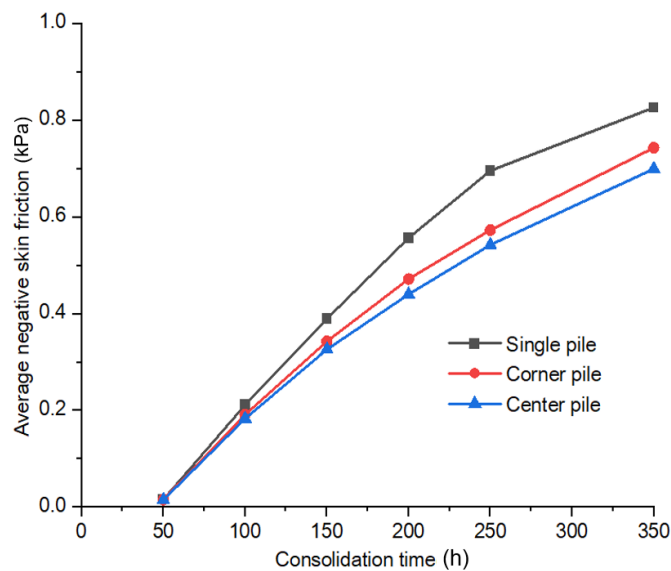


Figure 7. Influence of pile group effect on negative skin friction

Figure 7 illustrates the influence of the pile group effect on the average negative skin friction along the pile shaft. After 350 h of surcharge loading, the average negative skin friction of the single pile is 0.827 kPa, whereas the average negative skin friction values for the corner and center piles are 0.744 kPa and 0.7 kPa, respectively. The pile group effect coefficients for the average negative skin friction of the corner and center piles are 0.9 and 0.85, respectively. The pile group effect can effectively mitigate the development of negative skin friction on the pile foundation. Furthermore, because the center pile within the group is influenced by adjacent piles on both sides, it experiences a more pronounced pile group effect than the corner pile, resulting in a correspondingly smaller negative skin friction. The significant differences between the corner piles and the center pile in terms of displacement, axial force, and negative skin friction are primarily attributed to the shielding effect and stress diffusion mechanisms within the pile group. As the surcharge-induced additional stress propagates into the pile group, the peripheral corner piles bear the brunt, intercepting a substantial portion of the horizontal squeezing force and vertical effective stress. In contrast, the center pile is surrounded by adjacent piles, whose high rigidity restricts the free displacement of the soil between the piles, thereby providing pronounced “shielding” protection to the center pile. During this process, the stress attenuates drastically as it diffuses inward, leading to a substantial reduction in the additional stress ultimately transmitted to the center pile. Consequently, the dragload and displacement responses experienced by the center pile are noticeably weaker than those of the peripheral corner piles.

3.2 Analysis of the Influence of Isolation Pile Spacing

The pile-soil displacement characteristics and pile shaft load-bearing characteristics of the pile group under the condition without isolation piles (Test 2) and conditions with varying isolation pile spacings (Tests 3–5) were comparatively analyzed to investigate the influence of isolation pile spacing on the isolation effect of negative skin friction in pile groups under surcharge loading. Because the forces and displacements of the center pile are more significantly affected by the pile group effect, the center pile of the group was selected as the comparative reference pile.

3.2.1 Influence on the pile-head displacement of the center pile in the group

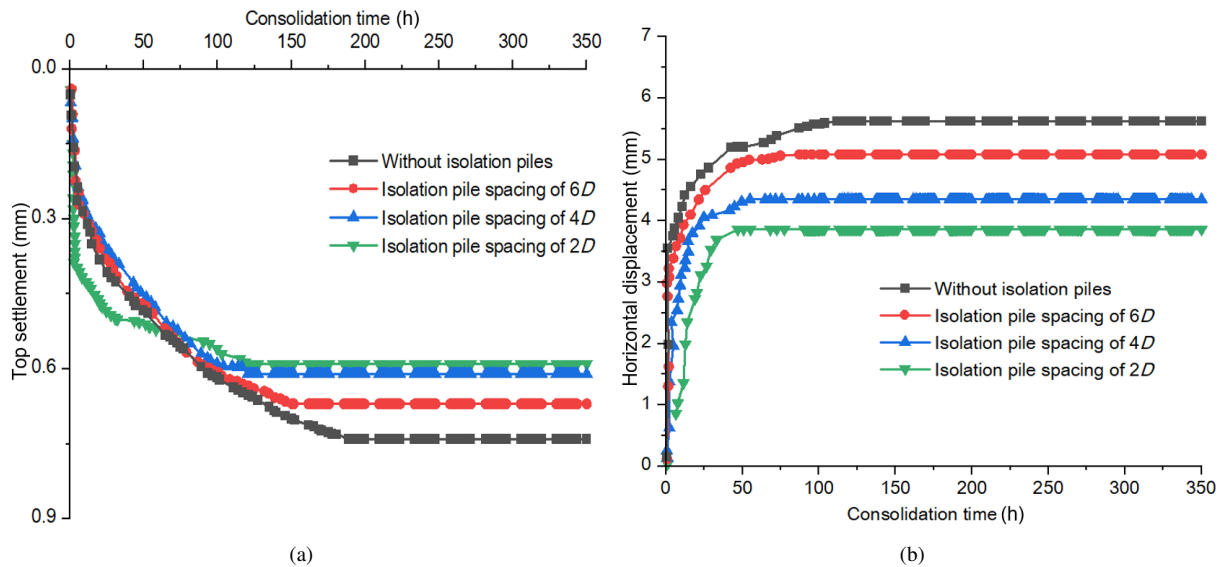


Figure 8. Influence of isolation pile spacing on pile top displacement: (a) Pile top settlement; (b) Pile horizontal displacement

The pile-head settlement curves for the engineering piles in each test group are presented in Figure 8a. At a consolidation time of 350 h, the pile-head settlement of the center engineering pile in the group without isolation piles is 0.74 mm. When isolation piles with a spacing of $6D$ are installed, the pile-head settlement of the engineering pile is reduced by 9.46% compared to the condition without isolation piles, indicating that the installation of isolation piles can effectively reduce the pile-head settlement of engineering piles. When the isolation pile spacing is reduced from $6D$ to $2D$, the reduction rate of pile-head settlement increases from 9.46% to 20.27%. Decreasing the isolation pile spacing can effectively enhance the protective effect on the pile-head settlement of engineering piles. This is because the robust load-bearing system of the isolation piles can longitudinally dissipate the sustained friction, intercepting the vertical deformation between the interior and exterior of the isolation piles, and thereby reducing the settlement of the soil and structures across the isolation boundary. Furthermore, following the installation of isolation piles,

the pile-head settlement of the engineering pile exhibits a relatively slower growth rate compared to the condition without isolation piles; the settlement only tends to stabilize after approximately 175 h. This demonstrates that while the isolation piles block the surcharge-induced stress, they also prolong the response time of the pile foundation behind them.

Figure 8b illustrates the variation of the horizontal displacement of the engineering pile with consolidation time under each test condition. When the isolation pile spacing is reduced from $6D$ to $2D$, the reduction rate of the horizontal displacement of the engineering pile increases from 9.61% to 31.32%, compared to the condition without isolation piles. Because isolation piles can intercept the lateral earth pressure induced by surcharge loading, installing them can effectively reduce the horizontal displacement of engineering piles. Meanwhile, as the isolation pile spacing decreases, the additional lateral pressure generated by the surcharge is blocked by the more densely arranged isolation piles; consequently, the horizontal displacement response of the engineering pile becomes smaller, resulting in a stronger mitigation effect on the horizontal displacement.

3.2.2 Influence on the pile shaft forces of the center pile in the group

(i) Axial force and bending moment

Figure 9a illustrates the variation of the engineering pile's axial force with depth under different isolation pile spacings. As the relative depth increases, the pile shaft axial force gradually increases, reaches its maximum value at a relative depth of 0.9, and subsequently begins to decrease. When isolation piles with a spacing of $6D$ are installed, the maximum axial force of the pile shaft is 55.1 N. Compared to the maximum axial force of 59.6 N under the condition without isolation piles, the axial force is reduced by 7.55%. When the isolation pile spacing is further reduced to $4D$ and $2D$, the reduction rate of the maximum axial force of the engineering pile increases to 15.6% and 20.13%, respectively. The smaller the isolation pile spacing, the more pronounced the reduction effect on the engineering pile's axial force. As the isolation pile spacing decreases, the barrier effect of the isolation piles against the surcharge-induced additional stress is enhanced. This reduces the soil settlement around the engineering piles and lowers the normal stress on the pile shaft, which in turn weakens the effect of negative skin friction along the pile shaft and ultimately decreases the pile shaft axial force.

Figure 9b presents the distribution of the engineering pile's bending moment with depth under different isolation pile spacings. The maximum bending moment of the pile shaft occurs approximately in the middle of the embedded section and then decreases as the relative depth increases. As the isolation pile spacing decreases from $6D$ to $4D$ and $2D$, the maximum bending moment of the engineering pile is reduced by 6.89%, 11.54%, and 19.37%, respectively, compared to the condition without isolation piles. This indicates that installing isolation piles can effectively reduce the bending moment of the engineering piles, and the mitigation effect strengthens as the pile spacing decreases. A smaller isolation pile spacing forms a denser barrier layer, which enhances the isolation effect against horizontal pressure and reduces the horizontal earth pressure acting on the sides of the engineering piles. Since the bending moment of the engineering pile shaft is induced by the horizontal earth pressure, a smaller isolation pile spacing consequently results in a smaller maximum bending moment.

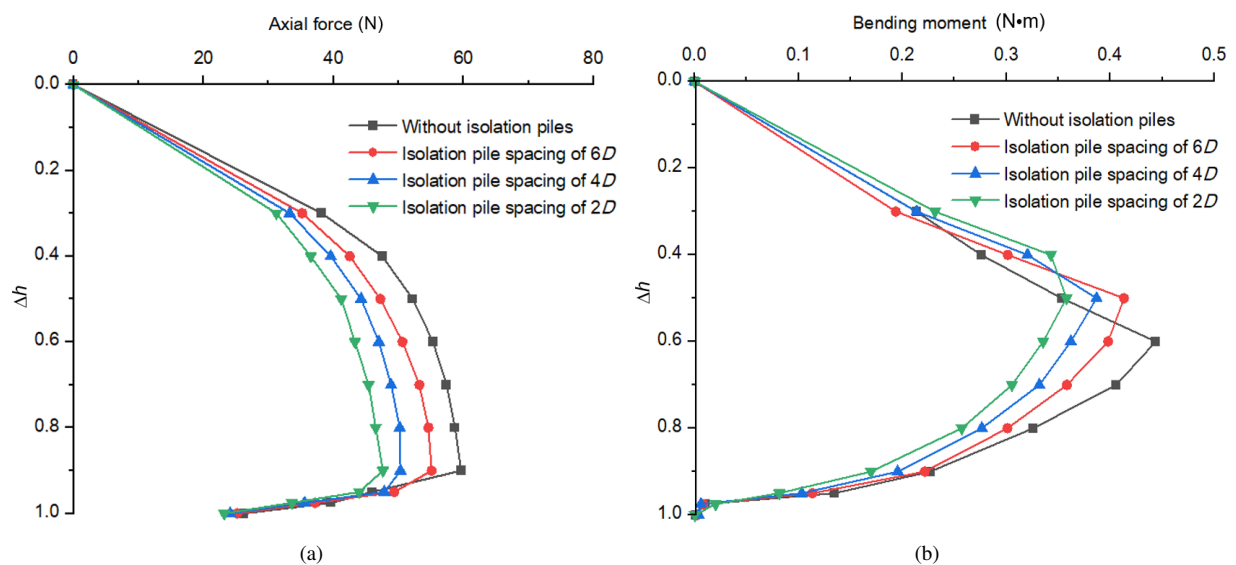


Figure 9. Influence of isolation pile spacing on axial force and bending moment: (a) Axial force profile; (b) Bending moment profile

(ii) Neutral plane

The location of the maximum axial force in Figure 9a corresponds to the position of the neutral plane. Under the conditions with different isolation pile spacings, the neutral plane is consistently located at 0.9 times the pile embedment depth. This aligns with the typical distribution characteristics of the neutral plane for end-bearing piles (where the pile tip rests in the sand layer).

3.2.3 Influence on the average negative skin friction of the center pile in the group

The development of negative skin friction along the pile shaft with soil consolidation is illustrated in Figure 10. The negative skin friction of the engineering pile increases with consolidation time, with the growth rate being initially rapid and subsequently slowing down. Following the installation of isolation piles, both the pile-head settlement and the horizontal earth pressure exerted on the engineering pile shaft are reduced; consequently, the average negative skin friction in the section above the neutral plane decreases accordingly. After 350 h of consolidation, the average negative skin friction of the engineering pile decreases as the isolation pile spacing becomes smaller. For isolation pile spacings of $6D$, $4D$, and $2D$, the reduction rates of the average negative skin friction on the engineering pile—compared to the condition without isolation piles—are 7.55%, 15.6%, and 20.13%, respectively. When the isolation pile spacing is reduced from $6D$ to $4D$, the isolation effect improves by 8.05%. However, when the spacing is further reduced from $4D$ to $2D$, the isolation effect only improves by 4.53%. This indicates that once the isolation pile spacing is reduced to $4D$, further decreasing the spacing no longer yields a significant enhancement in the isolation effect, as the rate of improvement diminishes. As the isolation pile spacing decreases, the stress influence zones of adjacent piles overlap significantly, resulting in strong group interaction effects. During this process, the majority of the additional stress undergoes redistribution and is concentrated on the row of isolation piles. When the spacing is sufficiently small, the lateral and vertical deformations of the soil between the piles are heavily restrained to their limits, and the system's capacity for stress redistribution approaches saturation. Consequently, the remaining additional stress available for further interception becomes minimal, leading to a notably diminished marginal protective benefit when the pile spacing is further reduced.

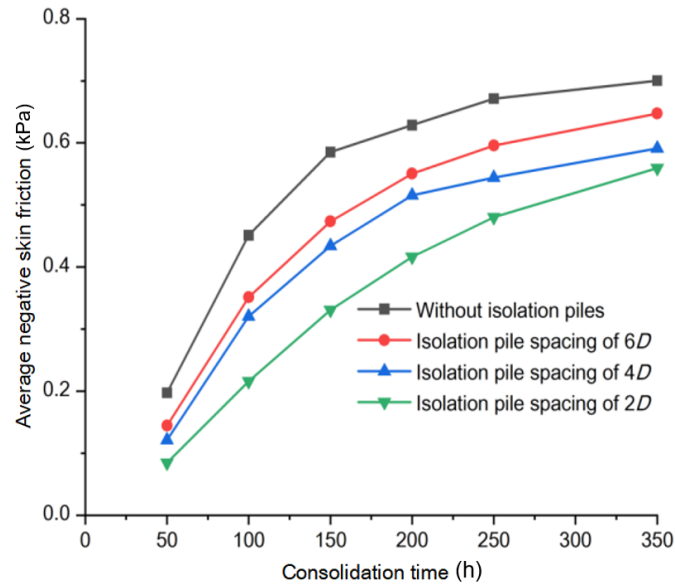


Figure 10. Influence of isolation pile spacing on the average negative skin friction of the engineering pile

4 Conclusions

Based on the issues of the pile group effect of surcharge-induced negative skin friction and the influence of isolation pile spacing on isolation efficacy, this study conducted a series of physical model tests and mechanistic investigations. The main conclusions are as follows:

(i) Pile foundations subjected to surcharge loading experience not only vertical negative skin friction but also horizontal loading, which induces bending moments and horizontal deflections in the pile shaft. The forces and displacements of the pile shaft increase with consolidation time. Because they are end-bearing piles, the maximum axial force occurs near the pile tip, while the maximum bending moment appears near the middle of the pile shaft.

(ii) The pile group effect significantly influences the forces and displacements of the pile shaft. It can mitigate the effect of negative skin friction on the pile foundation. Compared to a single pile, the average negative skin friction of the corner piles and center piles within the group is reduced by 10.04% and 15.36%, respectively. Because the

corner pile experiences a weaker shielding effect, its pile group effect coefficient is greater than that of the center pile.

(iii) Isolation piles can effectively reduce the negative skin friction of pile groups under surcharge loading. Under the present test conditions, installing isolation piles with a spacing of $2D$ provides the best protective effect for the engineering piles. Compared to the condition without isolation piles, the pile-head settlement, horizontal displacement, pile shaft axial force, bending moment, and average negative skin friction are reduced by 20.27%, 31.32%, 20.13%, 19.37%, and 20.13%, respectively.

(iv) The mitigation effect of isolation piles on the surcharge-induced negative skin friction of engineering piles increases as the isolation pile spacing decreases. When the isolation pile spacings are $6D$, $4D$, and $2D$, the average negative skin friction is reduced by 9.2%, 14.91%, and 20.13%, respectively, compared to the condition without isolation piles.

(v) In practical engineering, although reducing the isolation pile spacing can effectively decrease the negative skin friction on engineering piles, as the spacing is further reduced, the restriction of the isolation system on soil deformation approaches its limit, and the growth rate of the mitigation efficacy gradually slows down. When the isolation pile spacing is reduced from $6D$ to $4D$ and from $4D$ to $2D$, the mitigation effect on negative skin friction improves by 8.05% and 4.53%, respectively, indicating a diminishing marginal improvement. Therefore, in practical applications, mechanical performance and economic efficiency should be comprehensively evaluated to select the optimal isolation spacing.

Author Contributions

Conceptualization, C.S.C. and J.B.K.; methodology, C.S.C.; formal analysis, Y.W.; investigation, T.H.; resources, Q.W.Y.; data curation, P.L.L.; writing—original draft preparation, J.B.K.; writing—review and editing, T.H.; visualization, Y.W.; supervision, Q.W.Y.; project administration, P.L.L.; funding acquisition, C.S.C. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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